Multi-Eigenvalue Rosetta Stone for the Staircase Bott–Samelson Element: Palindromy via Bar-Involution

Verification at $n \leq 6$; integrality and palindromy proved; positivity open

Clio

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Abstract

Let $\Pi_q = \prod_{j=1}^{\binom{n}{2}}(1+T_{i_j})$ be the staircase Bott–Samelson element of the Iwahori–Hecke algebra $H_q(S_n)$, with $\mathbf{i} = (s_1)(s_2s_1)(s_3s_2s_1)\cdots$ a reduced word for $w_0 \in S_n$. For each irreducible V_λ with $r := \dim V_\lambda^H \geq 1$ (where $H = \langle s_1, s_3, \ldots\rangle$), the operator $\Pi_q|_{V_\lambda}$ has rank r; let $e_\lambda^{(1)}, \ldots, e_\lambda^{(r)} \in \overline{\mathbb{Q}(q)}$ be its nonzero eigenvalues.

The *Multi-Eigenvalue Rosetta Stone* conjecture asserts that for every $k \in \{1, \ldots, r\}$, the elementary symmetric function $\sigma_k(e_\lambda^{(1)}, \ldots, e_\lambda^{(r)})$ factors as $q^{a_{\lambda,k}}(1+q)^{b_{\lambda,k}}P_{\lambda,k}(q)$ with $P_{\lambda,k} \in \mathbb{Z}_{\geq 0}[q]$ palindromic and $P_{\lambda,k}(0) = 1$.

We prove three of the four ingredients and verify the full conjecture computationally for $n \leq 6$.

- **Integrality** (Theorem 3): each $\sigma_k \in \mathbb{Z}[q]$.
- **Palindromy** (Theorem 6): $\sigma_k(q^{-1}) = q^{-kN}\sigma_k(q)$ with $N = \binom{n}{2}$, via the bar-involution $\overline{b_s} = q^{-1}b_s$ transported through exterior powers.
- **Positivity on** $q > 0$ (Theorem 9): each $\sigma_k(q) > 0$ for real $q > 0$, via the eigenvalue-positivity theorem.
- **Non-negativity of integer coefficients**: open. Verified at $n \leq 6$ for all (λ, k).

We also exhibit a new datum at $n = 6$: the trace of Π_q on $V_{(3,2,1)}$ is divisible by $(1+q)^1$ only — not by $(1+q)^{\lfloor n/2 \rfloor} = (1+q)^3$. This refutes any direct generalization of the trace-level $(1+q)^{\lfloor n/2 \rfloor}$ result (proved for $n \leq 5$ in [3]) to $n \geq 6$.

1 Setup

Let $H_q(S_n)$ denote the Iwahori–Hecke algebra of S_n over $\mathbb{Z}[q, q^{-1}]$ with generators $T_1, \ldots, T_{n-1}$ and relations

$$T_i^2 = (q-1)T_i + q, \qquad T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \qquad T_iT_j = T_jT_i \ (|i-j| \geq 2).$$

Write $b_s := 1 + T_s$. Then $b_s^2 = (1+q)b_s$ (Lemma 1 below). Fix the staircase reduced expression

$$\mathbf{i} = (s_1)(s_2s_1)(s_3s_2s_1)\cdots(s_{n-1}\cdots s_1), \qquad \ell(\mathbf{i}) = N := \binom{n}{2},$$

and define the staircase Bott–Samelson element

$$\Pi_q := \prod_{j=1}^{N} b_{i_j} = \prod_{j=1}^{N}(1+T_{i_j}) \ \in \ H_q(S_n).$$

Let $H := \langle s_1, s_3, s_5, \ldots \rangle \cong (S_2)^{\lfloor n/2 \rfloor}$ be the parabolic generated by the odd Coxeter generators, and write $P_H = \prod_{i \in \{1,3,5,\ldots\} \cap [1,n-1]} b_i$. The H-symmetrizer $e_H := P_H/(1+q)^{\lfloor n/2 \rfloor}$ is an idempotent on any $H_q(S_n)$-module, and $\mathrm{im}(e_H|_V) = V^H$.

The operator $\Pi_q|_{V_\lambda}$ has rank $r_\lambda := \dim V_\lambda^H$ (proved for every λ in [1]). Let $e_\lambda^{(1)}, \ldots, e_\lambda^{(r_\lambda)} \in \overline{\mathbb{Q}(q)}$ denote its nonzero eigenvalues and

$$\sigma_k := \sigma_k(e_\lambda^{(1)}, \ldots, e_\lambda^{(r_\lambda)})$$

the kth elementary symmetric function, $1 \le k \le r_\lambda$.

Lemma 1. *$b_s^2 = (1+q)b_s$; hence $P_H^2 = (1+q)^{\lfloor n/2 \rfloor} P_H$ and e_H is idempotent.*

Proof. $(1+T_s)^2 = 1 + 2T_s + T_s^2 = 1 + 2T_s + (q-1)T_s + q = (1+q)(1+T_s)$. Odd generators pairwise commute, so $P_H^2 = \prod_i (1+T_i)^2 = (1+q)^s P_H$ where $s = \lfloor n/2 \rfloor$. □

2 The Multi-Eigenvalue Rosetta Stone Conjecture

Conjecture 2 (Multi-Eigenvalue Rosetta Stone)**.** For every $n \ge 1$ and every irreducible V_λ of S_n with $r_\lambda := \dim V_\lambda^H \ge 1$, and for every $k \in \{1, \ldots, r_\lambda\}$, there exist integers $a_{\lambda,k}, b_{\lambda,k} \ge 0$ and a polynomial $P_{\lambda,k} \in \mathbb{Z}_{\ge 0}[q]$ with $P_{\lambda,k}(0) = 1$ and $P_{\lambda,k}$ palindromic, such that

$$\sigma_k = q^{a_{\lambda,k}} (1+q)^{b_{\lambda,k}} P_{\lambda,k}(q). \tag{1}$$

The conjecture decomposes into four structural claims:

(I) **Integrality**: $\sigma_k \in \mathbb{Z}[q]$.

(II) **Palindromy**: after factoring out the largest q^a, the remainder σ_k/q^a is a palindromic polynomial.

(III) **Non-negativity**: all coefficients of σ_k are non-negative.

(IV) **Normalization**: the lowest-order (and equivalently highest-order) nonzero coefficient of σ_k is 1.

Given (I)–(IV), (1) follows: set $a = \min\{j : \sigma_k[q^j] \ne 0\}$, factor out the largest $(1+q)^b$ dividing σ_k/q^a (palindromy divided by the palindrome $(1+q)^b$ remains palindromic [4, standard]), and observe $P(0) = 1$ by (IV).

3 Integrality

Theorem 3 (Integrality)**.** *For every $\lambda \vdash n$ and every $k \ge 0$, $\sigma_k(e_\lambda^{(1)}, \ldots, e_\lambda^{(r_\lambda)}) \in \mathbb{Z}[q]$.*

Proof. The irreducible $H_q(S_n)$-module V_λ admits an integral form: Dipper–James show that the cell module (Specht module) for the Kazhdan–Lusztig left cell labeled by λ is a free $\mathbb{Z}[q, q^{-1}]$-module on which $H_q(S_n)$ acts by matrices in $\mathrm{Mat}_{d \times d}(\mathbb{Z}[q, q^{-1}])$ (with $d = \dim V_\lambda$). In fact, taking the integral Kazhdan–Lusztig basis, T_i and hence $b_i = 1 + T_i$ act by matrices with entries in $\mathbb{Z}_{\ge 0}[q]$ (W-graph positivity, provable for type A classically and in general via Elias–Williamson).

The characteristic polynomial of any such matrix $M \in \mathrm{Mat}_d(\mathbb{Z}[q, q^{-1}])$ lies in $\mathbb{Z}[q, q^{-1}][x]$. For $M = \Pi_q|_{V_\lambda}$ the matrix entries are in $\mathbb{Z}_{\ge 0}[q]$ (product of matrices with entries in $\mathbb{Z}_{\ge 0}[q]$), so $\det(xI - M) \in \mathbb{Z}[q][x]$.

The nontrivial factor of $\det(xI - M)$ is $\tilde{\chi}_\lambda(x, q) = \prod_{i=1}^{r_\lambda} (x - e_\lambda^{(i)})$, and its coefficients are (signed) elementary symmetric functions of the $e_\lambda^{(i)}$. Each coefficient of $\det(xI - M)$ lies in $\mathbb{Z}[q]$; hence $\sigma_k \in \mathbb{Z}[q]$. □

4 Palindromy via Bar-Involution and Exterior Powers

The key structural fact is that Π_q is self-bar-dual up to a power of q^{-1}; this transports through exterior powers to give palindromy of each σ_k.

Lemma 4 (Bar-transform of b_s)**.** *The bar involution* $\bar{\cdot} : H_q(S_n) \to H_q(S_n)$ *defined by* $\overline{q} = q^{-1}$, $\overline{T_w} = T_{w^{-1}}^{-1}$ *satisfies* $\overline{b_s} = q^{-1} b_s$.

Proof. From $T_s^2 = (q-1)T_s + q$ we get $T_s^{-1} = q^{-1}(T_s - (q-1)) = q^{-1}T_s + q^{-1} - 1$. Therefore

$$\overline{b_s} = 1 + \overline{T_s} = 1 + T_s^{-1} = q^{-1}T_s + q^{-1} = q^{-1}(T_s + 1) = q^{-1}b_s. \qquad \square$$

Lemma 5 (Bar-transform of Π_q)**.** $\overline{\Pi_q} = q^{-N}\Pi_q$, *with* $N = \binom{n}{2}$.

Proof. The bar involution is a $\mathbb{Z}$-algebra automorphism of $H_q(S_n)$. Hence $\overline{\Pi_q} = \overline{\prod_j b_{i_j}} = \prod_j \overline{b_{i_j}} = \prod_j (q^{-1}b_{i_j}) = q^{-N}\prod_j b_{i_j} = q^{-N}\Pi_q$. $\square$

Theorem 6 (Palindromy)**.** *For every* $\lambda \vdash n$ *with* $r_\lambda \geq 1$ *and every* $k \in \{1, \ldots, r_\lambda\}$,

$$\sigma_k(q^{-1}) = q^{-kN}\,\sigma_k(q). \tag{2}$$

Equivalently, writing σ_k *as a Laurent polynomial in* q*, the coefficient of* q^j *equals the coefficient of* q^{kN-j} *for every* $j \in \mathbb{Z}$.

Proof. Let M_λ be the matrix of Π_q acting on V_λ in the integral Kazhdan–Lusztig basis (Dipper–James cell basis). This basis is bar-invariant, so the bar involution of an entry $(M_\lambda)_{ij} \in \mathbb{Z}[q]$ is $(M_\lambda)_{ij}(q^{-1})$. Combined with Lemma 5,

$$M_\lambda(q^{-1}) = q^{-N}\,M_\lambda(q). \tag{3}$$

Fix $k \in \{1, \ldots, r_\lambda\}$. The kth exterior power of a linear operator is a functor on matrices; it scales as $\Lambda^k(cM) = c^k\Lambda^k(M)$ for any scalar c (since $\Lambda^k(cM)$ acts on $v_1 \wedge \cdots \wedge v_k \mapsto cMv_1 \wedge \cdots \wedge cMv_k = c^k\Lambda^k(M)(v_1 \wedge \cdots \wedge v_k)$). Applying Λ^k to (3):

$$\Lambda^k M_\lambda(q^{-1}) = q^{-kN}\,\Lambda^k M_\lambda(q).$$

Taking traces and using $\sigma_k = \mathrm{tr}(\Lambda^k M_\lambda) = \mathrm{tr}(\Lambda^k \Pi_q|_{V_\lambda})$ (basis-independent):

$$\sigma_k(q^{-1}) = q^{-kN}\,\sigma_k(q). \qquad \square$$

Corollary 7 (Palindromic structure)**.** *Write* $\sigma_k = \sum_{j\geq 0} c_{k,j}q^j$. *Then* $c_{k,j} = c_{k,kN-j}$ *for every* $j \geq 0$ *(where* $c_{k,j} = 0$ *if* $j > kN$*). In particular, if* a *is the smallest* j *with* $c_{k,j} \neq 0$ *and* D *is the largest, then* $a + D = kN$*, and* σ_k/q^a *is a palindromic polynomial of degree* $D - a = kN - 2a$.

Proof. Directly from the identity $c_{k,j} = c_{k,kN-j}$ in Theorem 6. $\square$

Corollary 8 (Divisibility by $(1+q)^b$ preserves palindromy)**.** *If* $\sigma_k \in \mathbb{Z}[q]$ *is palindromic in the sense of Corollary 7 and* $(1+q)^b$ *divides* σ_k/q^a *for the maximal such* a*, then the quotient* $\sigma_k/(q^a(1+q)^b)$ *is itself palindromic.*

Proof. $(1+q)^b$ is palindromic of degree b. If $R = (1+q)^b Q$ is palindromic of degree $D' = D - a$, then $q^{D'}R(q^{-1}) = R(q)$, i.e., $q^{D'}(1+q^{-1})^b Q(q^{-1}) = q^{D'-b}(1+q)^b Q(q^{-1}) = (1+q)^b Q(q)$; cancelling $(1+q)^b$ gives $Q(q) = q^{D'-b}Q(q^{-1})$, so Q is palindromic of degree $D' - b$. $\square$

5 Positivity on real $q > 0$

Theorem 9 (Positivity on $q > 0$)**.** *For every real $q > 0$, $\sigma_k(q) > 0$.*

Proof. The Eigenvalue Positivity Theorem [2] gives: every nonzero eigenvalue of $\Pi_q|_{V_\lambda}$, as an element of $\overline{\mathbb{Q}(q)}$, is positive for every specialization $q > 0$. Since each σ_k is the kth elementary symmetric function of positive real numbers (after specialization), $\sigma_k(q) > 0$. □

Theorem 9 rules out positive real roots of σ_k (viewed as a polynomial in q). Combined with palindromy, this rules out any root pair $\{q_0, q_0^{-1}\}$ of positive real roots. However, palindromy plus positivity on $q > 0$ is *not* enough to conclude non-negativity of integer coefficients: the palindromic polynomial $q^4 - q^3 + q^2 - q + 1 = \Phi_{10}(q)$ is positive on $q > 0$ but has negative coefficients.

6 Computational verification for $n \leq 6$

We computed $\Pi_q|_{V_\lambda}$ in Young's seminormal (Hoefsmit) form for every irreducible V_λ of S_n for $n \in \{3, 4, 5, 6\}$, extracted the characteristic polynomial, factored out the zero eigenvalues, and decomposed each σ_k as $q^a(1+q)^b P(q)$. Table 1 records the factorization. In every case $P \in \mathbb{Z}_{\geq 0}[q]$ is palindromic with $P(0) = 1$.

Palindromy is verified algebraically (as a polynomial identity $\sigma_k(q) = q^{kN}\sigma_k(q^{-1})$) and agrees with the bar-involution proof of Theorem 6.

Table 1: Factorization $\sigma_k = q^a(1+q)^b P(q)$ for every irrep of S_n, $n \leq 6$, with $r_\lambda \geq 1$. P is palindromic, non-negative, $P(0) = 1$.

n	λ	k	(a, b)	$P(q)$
3	(3)	1	$(0, 3)$	1
3	$(2, 1)$	1	$(1, 1)$	1
4	(4)	1	$(0, 6)$	1
4	$(3, 1)$	1	$(1, 2)$	$q^2 + 4q + 1$
4	$(2, 2)$	1	$(2, 2)$	1
5	(5)	1	$(0, 10)$	1
5	$(4, 1)$	1	$(1, 4)$	$q^4 + 7q^3 + 17q^2 + 7q + 1$
5	$(4, 1)$	2	$(3, 12)$	$q^2 + 4q + 1$
5	$(3, 2)$	1	$(2, 2)$	$q^4 + 9q^3 + 19q^2 + 9q + 1$
5	$(3, 2)$	2	$(5, 8)$	$q^2 + 4q + 1$
5	$(3, 1, 1)$	1	$(3, 2)$	$q^2 + 4q + 1$
5	$(2, 2, 1)$	1	$(4, 2)$	1
6	(6)	1	$(0, 15)$	1
6	$(5, 1)$	1	$(1, 7)$	$q^6 + 9q^5 + 36q^4 + 70q^3 + 36q^2 + 9q + 1$
6	$(5, 1)$	2	$(3, 18)$	$q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$
6	$(4, 2)$	1	$(2, 3)$	$q^8 + 13q^7 + 79q^6 + 245q^5 + 365q^4 + 245q^3 + 79q^2 + 13q + 1$
6	$(4, 2)$	2	$(5, 10)$	$(q^2 + 4q + 1)(q^8 + 17q^7 + 103q^6 + 313q^5 + 473q^4 + \ldots)$
6	$(4, 2)$	3	$(9, 21)$	$(q^2 + 4q + 1)^3$
6	$(4, 1, 1)$	1	$(3, 3)$	$q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$

Table 1 – continued

n	λ	k	(a,b)	$P(q)$
6	$(3,3)$	1	$(3,3)$	$q^6+12q^5+52q^4+88q^3+52q^2+12q+1$
6	$(3,2,1)$	1	$(4,1)$	$(q^2+4q+1)(q^4+11q^3+21q^2+11q+1)$
6	$(3,2,1)$	2	$(9,2)$	$q^{10}+16q^9+\cdots+16q+1$
6	$(2,2,2)$	1	$(6,3)$	1

Remark 10 (A new datum at $n=6$)**.** At $n=6$, $\lambda=(3,2,1)$, $k=1$: the trace $\mathrm{tr}_{V_{(3,2,1)}}(\Pi_q)=\sigma_1$ is divisible by $(1+q)^1$ only, not by $(1+q)^3=(1+q)^{\lfloor n/2\rfloor}$. This refutes the naive generalization of the result [3] (which established $(1+q)^{\lfloor n/2\rfloor} \mid \mathrm{tr}_{V_\lambda}(\Pi_q)$ for $n\le 5$) to arbitrary n. In particular, the per-eigenvalue $(1+q)^{\lfloor n/2\rfloor}$-divisibility conjecture fails at $n=6, \lambda=(3,2,1)$, since $\sigma_1=\sum e_i$ divisible by $(1+q)^{\lfloor n/2\rfloor}$ would follow from per-eigenvalue divisibility. Conjecture 2 remains consistent: it only requires $b_{\lambda,k}\ge 0$, which is satisfied.

Remark 11 (The universal palindrome q^2+4q+1)**.** Two separate phenomena feature the palindrome q^2+4q+1:

- $P_{(3,1),n=4,k=1}=q^2+4q+1$ (the original "Rosetta Stone" palindrome at $n=4$).
- It appears as a factor of σ_r at every rank-r irrep we have checked. At $n=6, \lambda=(4,2), r=3$: $P_{(4,2),k=3}=(q^2+4q+1)^3$.

A branching-rule explanation would identify (q^2+4q+1) as the "atomic" palindrome associated to $V_{(3,1)}$. We did not succeed in formalizing such a branching rule: the obvious restriction $S_n\to S_{n-1}$ is incompatible with the subgroup $H=\langle s_1,s_3,\ldots\rangle$, since the top odd generator lies in $S_n\setminus S_{n-1}$.

7 Gaps and open problems

Conjecture 2 reduces (given integrality, palindromy, and positivity on $q>0$) to:

(G1) **Non-negativity of integer coefficients.** Prove $\sigma_k\in\mathbb{Z}_{\ge 0}[q]$ for every (λ,k).

(G2) **Normalization.** Prove the smallest nonzero coefficient of σ_k is 1.

Attempted routes for (G1) and why they stalled

Route 1: Direct KL positivity. The matrix of Π_q in the KL basis has entries in $\mathbb{Z}_{\ge 0}[q]$ (W-graph positivity). The trace σ_1 is thus in $\mathbb{Z}_{\ge 0}[q]$, settling (G1) for $k=1$. However for $k\ge 2$, $\sigma_k=\sum_{|S|=k}\det(M[S,S])$ is a sum of $k\times k$ principal minors of M, and these determinants can have mixed signs even when M has non-negative entries. The direct sum is not manifestly positive.

Route 2: Lindström–Gessel–Viennot. Using $\Pi_q=b_{i_1}\cdots b_{i_N}$ in the KL basis, a telescoping application of Cauchy–Binet gives

$$\sigma_k \;=\; \sum_{S_0=S_N,\,S_1,\ldots,S_{N-1}}\;\prod_{j=1}^{N}\det\big(b_{i_j}|_{S_{j-1},S_j}\big),$$

where the sum runs over all closed sequences of k-subsets. If each $b_{i_j}|_{V_\lambda}$ were a *totally non-negative* matrix (TNN), i.e., if every minor $\det(b_{i_j}|_{S,T})$ were non-negative, this would give $\sigma_k\ge 0$ term-by-term.

We tested TNN-ness for $b_{s_1}|_{V_{(2,1)}}$ and $b_{s_1}|_{V_{(3,1)}}$ in several natural bases (Kazhdan–Lusztig, Murphy standard, Hoefsmit seminormal) and found: the diagonal seminormal b_{s_1} is TNN trivially (eigenvalue matrix); in the KL basis the entries are non-negative but the 2-minors are not in general. So TNN-ness does *not* hold in the natural bases, and the LGV route stalls.

Route 3: Verdier duality / Soergel decomposition. In the Elias–Williamson Hecke category, Π_q corresponds to the Bott–Samelson object $BS(\mathbf{i})$, which decomposes as a direct sum of indecomposable Soergel bimodules $BS(\mathbf{i}) \simeq \bigoplus_w m_w(\mathbf{i}, q)B_w$ with Kazhdan–Lusztig multiplicities $m_w \in \mathbb{Z}_{\geq 0}[q, q^{-1}]$. The character of each B_w on V_λ is a polynomial in q. This would yield an explicit Schur-positive-type expansion of $\sigma_1 = \mathrm{tr}_{V_\lambda}(\Pi_q)$; for $k \geq 2$, $\sigma_k = \mathrm{tr}(\Lambda^k \Pi_q)$ does not come from an element of $H_q(S_n)$ (exterior power is not functorial on $H_q(S_n)$-modules), so the direct Soergel route applies to σ_1 only.

What would close (G2)

Normalization $P(0) = 1$ is equivalent to: the lowest nonzero coefficient of σ_k equals 1. From palindromy this coincides with the leading coefficient. The leading coefficient of σ_k is the q^{kN}-coefficient of σ_k, which by palindromy equals the constant term of σ_k/q^a for $a = kN - \deg(\sigma_k)$. Computation shows this constant term is 1 in all observed cases; a structural proof eludes us.

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References

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